

Input Views & Plücker Rays



Visual Geometry
Conditioner

VAE Encoder

PixelUnshuffle

Target Rays

WAN-Init Multi-View Diffusion

Dropped with probability
during training

Pixel-Aligned Concat

Conv3D Patch Embed

Timestep AdaLN

Self Attention

FFN

DiT Block $\times 40$

Linear & Unpatchify

1. Flow matching loss

$$\mathcal{L}_{\text{FM}} = \|V_{\theta,j} - U_j\|$$



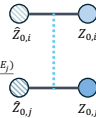
2. Point-track residual consistency

$$\hat{Z}_{0,j} = X_{\sigma,j} - \sigma V_{\theta,j}$$

$$E_j = \hat{Z}_{0,j} - Z_{0,j}$$

$$\mathcal{L}_{\text{PTRC}} = \frac{\sum_{c \in \mathcal{C}} \omega_c \bar{\rho}_\beta(E_i - E_j)}{\sum_{c \in \mathcal{C}} \omega_c}$$

$$\mathcal{L} = \mathcal{L}_{\text{FM}} + \lambda_{\text{PTRC}} \mathcal{L}_{\text{PTRC}}$$



Target Views

VAE Decoder

 Routed visual geometry token

 Plücker map token

 Clean source latent

 Noisy target latent